

§ 173.185 Lithium cells and batteries.

As used in this section, *consignment* means one or more packages of hazardous materials accepted by an operator from one shipper at one time and at one address, receipted for in one lot and moving to one consignee at one destination address. *Equipment* means the device or apparatus for which the lithium cells or batteries will provide electrical power for its operation. *Lithium cell(s)* or *battery(ies)* includes both lithium metal and lithium ion chemistries. *Medical device* means an instrument, apparatus, implement, machine, contrivance, implant, or in vitro reagent, including any component, part, or accessory thereof, which is intended for use in the diagnosis of disease or other conditions, or in the cure, mitigation, treatment, or prevention of disease, of a person.

(a) **Classification.**

(1) Each lithium cell or battery must be of the type proven to meet the criteria in part III, sub-[section 38.3](#) of the UN Manual of Tests and Criteria (IBR; see [§ 171.7 of this subchapter](#)). Lithium cells and batteries are subject to these tests regardless of whether the cells used to construct the battery are of a tested type. A single cell battery as defined in part III, sub-[section 38.3](#) of the UN Manual of Tests and Criteria is considered a “cell” and must be offered for transportation in accordance with the requirements for cells.

(i) Cells and batteries manufactured according to a type meeting the requirements of sub-[section 38.3](#) of the UN Manual of Tests and Criteria, Revision 3, Amendment 1 or any subsequent revision and amendment applicable at the date of the type testing may continue to be transported, unless otherwise provided in this subchapter.

(ii) Cell and battery types only meeting the requirements of the UN Manual of Tests and Criteria, Revision 3, are no longer valid. However, cells and batteries manufactured in conformity with such types before July 2003 may continue to be transported if all other applicable requirements are fulfilled.

(2) Each person who manufactures lithium cells or batteries must create a record of satisfactory completion of the testing (e.g. test report) required by this paragraph prior to offering the lithium cell or battery for transport and must:

(i) Maintain this record for as long as that design is offered for transportation and for one year thereafter; and

(ii) Make this record available to an authorized representative of the Federal, state or local government upon request.

(3) Each manufacturer and subsequent distributor of lithium cells or batteries, except for button cells installed in equipment (including circuit boards), manufactured on or after

January 1, 2008, must make a test summary available. The test summary must include the following elements:

- (i) Name of cell, battery, or product manufacturer, as applicable;
- (ii) Cell, battery, or product manufacturer's contact information to include address, telephone number, email address, and website for more information;
- (iii) Name of the test laboratory, to include address, telephone number, email address, and website for more information;
- (iv) A unique test report identification number;
- (v) Date of test report;
- (vi) Description of cell or battery to include at a minimum;
 - (A) Lithium ion or lithium metal cell or battery;
 - (B) Mass of cell or battery;
 - (C) Watt-hour rating, or lithium content;
 - (D) Physical description of the cell/battery; and
 - (E) Cell or battery model number or, alternatively, if the test summary is established for a product containing a cell or battery, the product model number.
- (vii) List of tests conducted and results (*i.e.*, pass/fail);
- (viii) Reference to assembled battery testing requirements (if applicable);
- (ix) Reference to the revised edition of the UN Manual of Tests and Criteria used and to amendments thereto, if any; and
- (x) Name and title of a responsible person as an indication of the validity of information provided.

(4) Except for cells or batteries meeting the requirements of [paragraph \(c\)](#) of this section, each lithium cell or battery must:

- (i) Incorporate a safety venting device or be designed to preclude a violent rupture under conditions normally incident to transport;
- (ii) Be equipped with means of preventing external short circuits; and
- (iii) Be equipped with a means of preventing dangerous reverse current flow (*e.g.*, diodes or fuses) if a battery contains cells, or a series of cells that are connected in parallel.

(5) Beginning May 10, 2024, each lithium ion battery must be marked with the Watt-hour rating on the outside case.

(b) *Packaging.*

(1) Each package offered for transportation containing lithium cells or batteries, including lithium cells or batteries packed with, or contained in, equipment, must meet all applicable requirements of [subpart B of this part](#).

(2) Lithium cells or batteries, including lithium cells or batteries packed with, or contained in, equipment, must be packaged in a manner to prevent:

- (i) Short circuits;
- (ii) Damage caused by shifting or placement within the package; and
- (iii) Accidental activation of the equipment.

(3) For packages containing lithium cells or batteries offered for transportation:

(i) The lithium cells or batteries must be placed in non-metallic inner packagings that completely enclose the cells or batteries, and separate the cells or batteries from contact with equipment, other devices, or electrically conductive materials (e.g., metal) in the packaging.

(ii) The inner packagings containing lithium cells or batteries must be placed in one of the following packagings meeting the requirements of part 178, subparts L and M, of this subchapter at the Packing Group II level:

(A) Metal (4A, 4B, 4N), wooden (4C1, 4C2, 4D, 4F), fiberboard (4G), or solid plastic (4H1, 4H2) box;

(B) Metal (1A2, 1B2, 1N2), plywood (1D), fiber (1G), or plastic (1H2) drum;

(C) Metal (3A2, 3B2) or plastic (3H2) jerrican.

(iii) When packed with equipment, lithium cells or batteries must:

(A) Be placed in inner packagings that completely enclose the cell or battery, then placed in a packaging of a type that meets the Packing Group II performance requirements as specified in [paragraph \(b\)\(3\)\(ii\)](#) of this section, and then placed with the equipment in a strong, rigid outer packaging; or

(B) Be placed in inner packagings that completely enclose the cell or battery, then placed with the equipment in a packaging of a type that meets the Packing Group II performance requirements as specified in [paragraph \(b\)\(3\)\(ii\)](#) of this section.

(C) For transportation by aircraft, the number of cells or batteries in each package is limited to the minimum number required to power the piece of equipment, plus two spare sets. A set of cells or batteries is the number of individual cells or batteries that are required to power each piece of equipment.

(4) When lithium cells or batteries are contained in equipment:

(i) The outer packaging, when used, must be constructed of suitable material of adequate strength and design in relation to the capacity and intended use of the packaging, unless the lithium cells or batteries are afforded equivalent protection by the equipment in which they are contained;

(ii) Equipment must be secured to prevent damage caused by shifting within the outer packaging and be packed so as to prevent accidental operation during transport;

(iii) Any spare lithium cells or batteries packed with the equipment must be packaged in accordance with [paragraph \(b\)\(3\)](#) of this section; and

(iv) For transportation by aircraft, where multiple pieces of equipment are packed in the same outer packaging, each piece of equipment must be packed to prevent contact with other equipment.

(5) Lithium cells or batteries that weigh 12 kg (26.5 pounds) or more and have a strong, impact-resistant outer casing, may be packed in strong outer packagings; in protective enclosures (for example, in fully enclosed or wooden slatted crates); or on pallets or other handling devices, instead of packages meeting the UN performance packaging requirements in [paragraphs \(b\)\(3\)\(ii\)](#) and [\(b\)\(3\)\(iii\)](#) of this section. Cells and batteries must be secured to prevent inadvertent shifting, and the terminals may not support the weight of other superimposed elements. Cells and batteries packaged in accordance with this paragraph may be transported by cargo-only aircraft if approved by the Associate Administrator.

(6) Except for transportation by aircraft, the following rigid large packagings are authorized for a single battery, and for a single item of equipment containing batteries, meeting provisions in [paragraphs \(b\)\(1\)](#) and [\(2\)](#) of this section and the requirements of part 178, subparts P and Q, of this subchapter at the Packing Group II level:

(i) Metal (50A, 50B, 50N) metal packagings must be fitted with an electrically non-conductive lining material (e.g., plastics) of adequate strength for the intended use;

(ii) Rigid plastic (50H);

(iii) Wooden (50C, 50D, 50F);

(iv) Rigid fiberboard (50G).

(7) For transportation by aircraft, lithium cells and batteries must not be packed in the same outer packaging with substances and articles of Class 1 (explosives) other than Division 1.4S, Division 2.1 (flammable gases), Class 3 (flammable liquids), Division 4.1 (flammable solids), or Division 5.1 (oxidizers).